ADS203TC - Signal and Image Processing

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Academy of AI and Advanced Technology, Xi'an Jiaotong - Liverpool University

September 6, 2026

Welcome and Module Overview

This module provides a foundational treatment of signal and image processing, treating the two-dimensional image as a natural generalisation of the one-dimensional signal.

What you will learn

- How to **represent** signals and images, and how to **transform** and **filter** them.
- The theory and the practical, computer-based skills together: every concept is implemented in Python.
- The mathematics that underpins machine perception, data preprocessing, and feature engineering.

Why this module matters

Signals and images are the raw input to nearly every modern AI system. This module gives you the tools to reason about that data, rather than treating it as a black box.

Course Philosophy and Structure

This curriculum provides a comprehensive understanding of digital signal processing across two fundamental domains: one-dimensional (1D) temporal signals and two-dimensional (2D) spatial signals (images). The primary pedagogical objective is to build a rigorous theoretical foundation and then apply it to solve practical problems.

Part I: 1D Signal Processing

1. Introduction to Signal Processing
2. Convolution and Correlation
3. Sampling and Quantization
4. The Discrete Fourier Transform (DFT)
5. The Fast Fourier Transform (FFT)
6. Reliable Signal Transmission: Error-Correcting Codes (ECC)

Part II: 2D Image Processing

7. Image Fundamentals and Point Processes
8. Spatial Domain Filtering
9. Frequency-Domain Image Filtering
10. Image Compression
11. Digital Signal and Image Watermarking
12. Morphological Image Processing

Week 13

Module review

Module Delivery Pattern

Weekly Structure

- **13 x 2-hour Lectures**
  - Introduction of core concepts and theories.
- **13 x 2-hour Lab Sessions**
  - Hands-on problem solving and implementation of concepts using Python.
- **Private study**
  - Time for reflection, background reading, and completing the assessments.

Total learning time is 150 hours: 26 hours of lectures, 26 hours of seminars and labs, and 98 hours of private study. Please refer to the Module Handbook and the timetable for the detailed schedule.

Synthesis of Acquired Knowledge

This module synthesizes foundational theory with computational practice, culminating in a versatile skill set applicable across scientific and industrial domains.

Main Outcome

Ability to develop a unified perspective on signal processing, recognizing that concepts such as convolution, filtering, and Fourier analysis are fundamental mathematical tools applicable to any domain, be it temporal or spatial.

Demonstrable Capabilities Post-Completion

- Translate ambiguous real-world problems (e.g., "audio signal cleaning") into precise signal processing workflows.
- Select the appropriate transform, filter type, or analytical technique based on signal characteristics and desired outcomes.
- Analyze results with a deep understanding of the underlying mathematical principles and potential artifacts, such as aliasing, spectral leakage, or filtering transients.

Module learning outcomes

These are the four key goals for you to achieve by the end of this module:

A. Demonstrate a sound understanding of signal processing and image processing knowledge in both the theoretical and practical aspects.

B. Determine the most appropriate sampling and filtering methodology according to application.

C. Apply basic techniques of digital signal and image processing to numerically solve given problems.

D. Apply standard image processing techniques, e.g., image enhancement, image transform, image compression, and morphological operations.

Assessments

Coursework 1 (50%)

- Learning outcomes A and B.
- Individual, project-based coursework.
- **Release Date:** 18 September 2026
- **Due Date:** Wednesday, 11 November 2026, 23:59

Coursework 2 – Project (50%)

- Learning outcomes C and D.
- Individual, project-based coursework.
- **Release Date:** 19 October 2026
- **Due Date:** Wednesday, 16 December 2026, 23:59

Artificial Intelligence

Substantive use of AI is permitted for the assessments, provided it is properly disclosed and attributed. See the assessment task sheets and module handbook for the exact scope.

Resit Assessment

Resit Coursework (100%)

- Assesses all learning outcomes (A, B, C, D).
- Individual coursework.
- **Release Date:** 20 July 2027
- **Due Date:** Friday, 6 August 2027, 23:59

Artificial Intelligence in This Module

Artificial intelligence is part of how this module is taught and assessed.

The AI Tutor

An AI Tutor is embedded on the Learning Mall page. It is a study aid for exploring ideas and asking questions. It does not replace the lectures, the notes, or the teaching team.

AI in the lab

The Week 1 lab asks you to generate solutions with AI, then verify them against the theory. Verification, not generation, is the skill being developed.

AI in assessment

AI may be used in the coursework within a defined scope. Every task sheet states exactly what is allowed and how it must be declared.

AI in Assessment: Three Permission Levels

Every assessment states one of three AI permission levels, taken from the University Academic Integrity Policy. The level is printed in the task sheet and the module handbook.

Level 1: No AI use

AI tools are strictly forbidden for the assignment.

Level 2: Non-substantive use

AI only for minor tasks such as grammar, formatting, or generic suggestions. A brief citation of the tool name and version is required.

Level 3: Substantive use

AI may generate content, code, or analysis, or help structure an argument. Full disclosure is required: tool name, version, access date, and a description of the AI's role.

This module permits substantive use of AI in its assessments.

Using AI Well and Within the Rules

The AI Tutor has limits

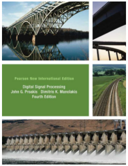
- **Not a primary source.** It can be wrong. The lecture notes and core texts are authoritative for this module.
- **Not an administrator.** It does not know deadlines, formats, or marks. The module handbook does.
- **Critical thinking is required.** Check every answer against your notes and the core texts.

Going beyond the permitted scope

Using AI beyond the stated scope is a breach of the Academic Integrity Policy. Penalties range from a reduced mark to zero for the task, and in serious cases zero for the module and disciplinary action.

Use AI to learn, not as a substitute for learning.

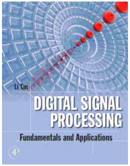
Recommended Textbooks



Primary – Signal Processing

Digital Signal Processing, 4th ed.

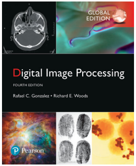
John G. Proakis & Dimitris K. Manolakis



Supplementary – Signal Processing

Digital Signal Processing: Fundamentals and Applications

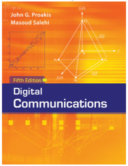
Li Tan



Primary – Image Processing

Digital Image Processing, 4th ed.

Rafael C. Gonzalez & Richard E. Woods



Week 6 – Error-Correcting Codes

Digital Communications, 5th ed.

John G. Proakis & Masoud Salehi

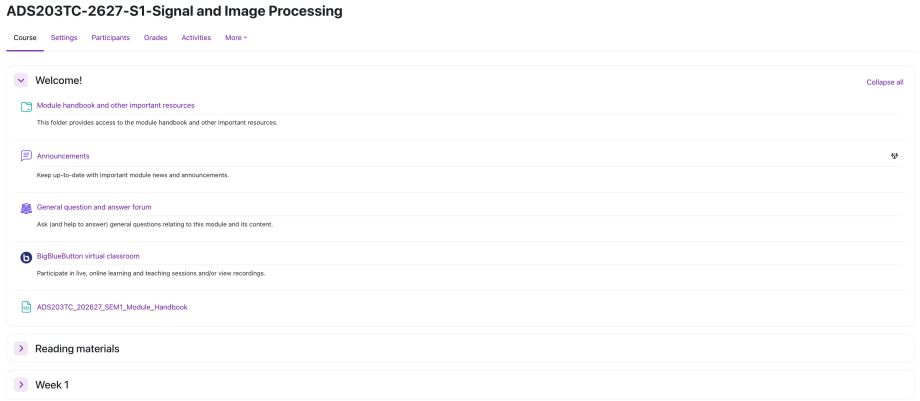
Full details, including the official optional texts, are in the module handbook.

Learning Mall

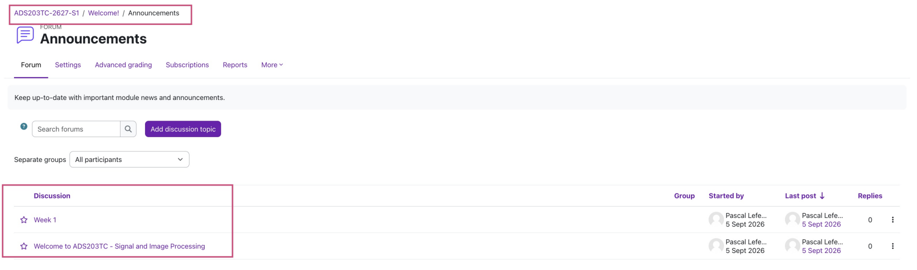
Our usual platform for module material (lectures, labs, module handbook, resources, FAQ, etc).

URL: https://core.xjtlu.edu.cn

Learning Mall Core



Announcements



Teaching Team

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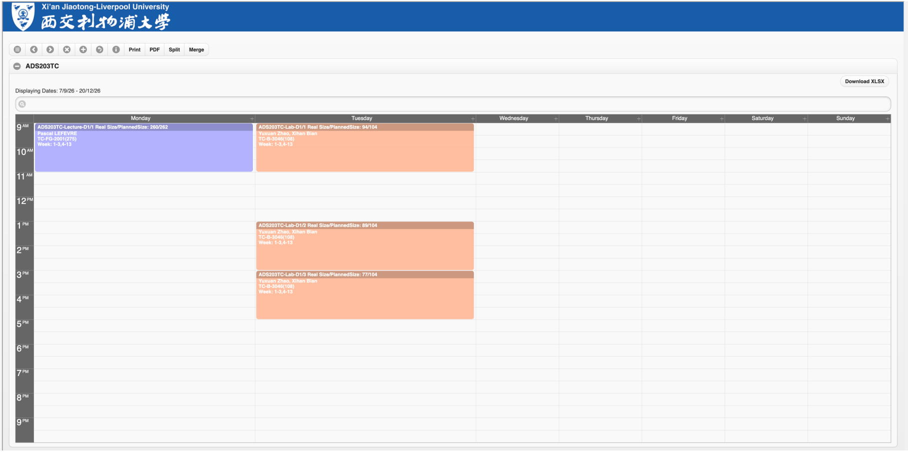
Office Hours

- Dr. Pascal Lefèvre: Mondays 1 pm - 3 pm; Tuesdays 1 pm - 3 pm, D-5012
- Dr. Xihan Bian: Tuesdays 1 pm - 3 pm; Wednesdays 2 pm - 4 pm, D-5013
- Dr. Yuxuan Zhao: Mondays 1 pm - 3 pm; Tuesdays 3 pm - 5 pm, D-5008

Teaching Assistant

Yihan Xin, yihan.xin26@student.xjtlu.edu.cn

Timetable – Semester 1 2026-2027



Biography

Module Leader

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Dr. Pascal Lefèvre

Dr. Pascal Lefèvre is an Assistant Professor in the Academy of AI and Advanced Technology. He received a Master's Degree in Cryptography and Computer Security from Bordeaux University, France, and then his Ph.D. from Poitiers University and XLIM Laboratory, France. He joined Shenzhen University as a Post-Doctoral researcher and later, worked in AI and Computer Vision in the tech industry. His past research involved multiple fields such as Cryptography, Error-Correcting Codes, Signal and Image Processing, Multimedia Security, and Artificial Intelligence. He actively works on Neural Network Compression and the applications of AI for Mathematics and AI for Science.

Biography

Co-Teacher

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Dr. Xihan Bian

Bian Xihan is a novel researcher with experience in Machine Learning, Robotics, Reinforcement Learning and Imitation Learning. He reviewed his Bachelors degree from the University of Texas at Austin. He then worked as an engineer ranging from high-speed computation and commutation systems to autonomous vehicles before earning his PhD with Dr.Simon Hadfield and Dr.Oscar Mendez in 2022. His research effort will continue in the field of robotics in combination with intelligent agents.

Biography

Co-Teacher

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Dr. Yuxuan Zhao

Dr Yuxuan Zhao is a lecturer at the Academy of AI and Advanced Technology (AIAT) at XJTLU Entrepreneur College (Taicang). He received his PhD in Computer Science from the University of Liverpool in 2022, and MSc in Computer Graphics, Vision and Imaging from University College London in 2017.